

# Learnix

## AI-Powered Inclusive Learning & Accessibility Ecosystem

### Hackathon Judge Presentation & Architecture Guide

### Executive Summary & Platform Mission

**Learnix** is an enterprise-grade, accessibility-first AI educational ecosystem designed to bridge learning barriers for students with visual, auditory, cognitive, and linguistic challenges. By combining multi-source document ingestion, optical character recognition (OCR), neural Machine Translation into scheduled Indian languages, multi-language Text-to-Speech (TTS) narration, and Retrieval-Augmented Generation (RAG) AI Tutoring, Learnix provides a unified platform where every student can learn in their preferred style and language.

### Key Impact Highlights & Standards Alignment

- **W3C WCAG 2.1 AAA Compliant:** Features OpenDyslexic typography, high contrast modes, focus rulers, and automated caption generation.
- **NEP 2020 Aligned:** Fulfills National Education Policy Clause 4.11 for mother-tongue instruction & Section 23 for NETF educational technology.
- **Digital India Bhashini Vision:** Breaks regional language barriers across 9 scheduled Indian languages (Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Bengali, English).

### Complete Technology Stack & Libraries Used

| Layer                    | Technologies & Libraries Used                                                                                               | Core Function & Purpose                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Frontend Framework       | React 18, Vite 5, Tailwind CSS, Lucide React, Recharts, React Router v6                                                     | Single Page Application (SPA), accessible glassmorphism UI, interactive charts, dyslexia font toggles.                    |
| Backend API Engine       | Python 3.11, FastAPI, Uvicorn ASGI, Pydantic, SQLAlchemy ORM                                                                | High-performance asynchronous REST API backend routing, data validation, and database ORM abstraction.                    |
| AI / LLM Services        | Google Gemini API (gemini-3.5-flash / gemini-2.0-flash / gemini-pro)                                                        | Document Q&A; RAG tutoring, intelligent content summarization, dynamic quiz generation, and accessibility audits.         |
| Translation & Speech     | deep-translator (Google Translate API engine), gTTS (Google Text-to-Speech)                                                 | Neural text translation across 9 Indian languages & native script MP3 audio narration + WebVTT captions.                  |
| OCR & Document Ingestion | PyMuPDF (fitz), pytesseract (Tesseract OCR), python-docx, python-pptx, Pillow (PIL), BeautifulSoup4, youtube-transcript-api | Text extraction from PDFs, DOCX, PPTX, image OCR processing, web scraping, and YouTube caption extraction.                |
| Database & Security      | PostgreSQL 15 (Production), SQLite (Fallback), Passlib (bcrypt), python-jose (JWT)                                          | Relational database storage, encrypted password hashing, JWT bearer token authorization, and user data privacy isolation. |

# Detailed Ecosystem Modules & Architecture

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## Module 1: Upload & Multi-Source Ingestion Center

- **Supported Input Sources:** PDF documents, Word (DOCX), PowerPoint (PPTX), Images (PNG, JPG), Web URLs, and YouTube Video URLs.
- **Multi-Engine Processing:** PyMuPDF parses digital PDFs; Tesseract OCR extracts text from scanned documents/slides; BeautifulSoup4 scrapes web articles; youtube-transcript-api fetches closed-captions.
- **Automated Subject Tagging:** Auto-classifies uploads into subjects (AI, Science, Mathematics, Literature, History) with initial readability metrics.

## Module 2: Multilingual Translation Center

- **9 Indic Languages Supported:** Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Bengali, and English.
- **Neural Machine Translation:** Powered by Google Gemini API & deep-translator for domain-specific terminology preservation.
- **Version Control:** Stores multiple translation iterations per document in the database, allowing instant switching between versions.

## Module 3: Accessibility Suite & Multi-Language TTS

- **Multi-Language Audio Narration:** Converts translated text into native-script MP3 audio using gTTS with speed control (0.75x to 1.5x) and voice gender options.
- **WebVTT Closed Captions:** Automatically generates synchronized subtitle files (.vtt) for screen reader accessibility.
- **W3C Reading Modes:** OpenDyslexic font spacing, High Contrast pure-black mode, Reading Ruler focus guide, and responsive text sizing.
- **Accessibility Compliance Suite:** Format-specific automated WCAG audits:
  - **YouTube Videos:** Caption Availability Check + Spoken Language Check.
  - **Images & Slides:** Color Contrast Analysis (Low contrast & unreadable combinations detection with WCAG recommendations) + OCR Check.
  - **PDF Documents:** OCR Selectability, Heading Structure Analysis (H1 missing, skipped levels), Language Detection, Font Accessibility.
  - **Web URLs:** Web text selectability, HTML heading structure, language detection, and typography.

## Module 4: Smart Quiz & Revision Generator

- **Content-Driven Generation:** Parses uploaded study content into 4 question formats: Multiple Choice (MCQs), True/False, Fill-in-the-Blanks, and Flashcards.
- **Multi-Language Quiz Translation:** Generates quiz questions directly in the user's chosen target language (e.g. Tamil or Hindi).
- **Interactive Assessment:** Live score calculation, instant feedback, explanation popups, and progress recording.

# Detailed Ecosystem Modules (Continued)

## Module 5: AI Study Assistant (Document Chat / RAG)

- **Gemini RAG Architecture:** Queries document context using Gemini API (gemini-3.5-flash) with Extractive QA semantic search fallback.
- **'Explain Simply' Mode:** Simplifies complex technical terms into beginner-friendly bullet points for neurodivergent learners.
- **On-the-Spot Answer Translation:** Translates answers into 9 Indian languages dynamically.

## Module 6: Learning Library & Resource Manager

- **Centralized Vault:** Organizes all uploaded documents, generated audio narrations, translations, and quizzes.
- **Search & Filtering:** Instant search by document title, subject, file format, or upload date.

## Module 7: Analytics & Learning Progress Dashboard

- **Interactive Charts:** Powered by Recharts to display weekly learning progress, accessibility scores, and document completion metrics.
- **Gamified Metrics:** Displays hours learned, audio listened, and quizzes completed.

## Module 8: Admin Control Panel & Diagnostics

- **System Diagnostics:** Monitors REST API health, CPU/RAM usage, database connection status, storage space, and active sessions.

## Module 9: Authentication, Profile & Data Privacy

- **Secure Database Auth:** User account signup and login using bcrypt hashed passwords stored in PostgreSQL/SQLite.
- **User Data Privacy Isolation:** Enforces strict user\_id database filtering so each user's history and documents are visible ONLY to them.
- **Session Logout:** Full JWT token cleanup and protected route navigation gates.

# Hackathon Judging Criteria Alignment Summary

| Judging Criteria                  | How Learnix Excels & Demonstrates Excellence                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Innovation & Technical Complexity | Integrates 5 AI/NLP engines (LLM RAG, Neural MT, gTTS, OCR, Web/YouTube scraping) into a seamless, reactive full-stack web application. |
| Social Impact & Inclusion         | Empowers neurodivergent students (dyslexia, ADHD), visually/auditory impaired learners, and regional language students across India.    |
| Execution & UI/UX Design          | Stunning modern glassmorphism design, high contrast toggles, OpenDyslexic font support, responsive layout, 0% latency UI transitions.   |
| Completeness & Real-World Utility | Fully functional end-to-end platform with database persistence, JWT authentication, fallback mechanisms, and zero mock place-holders.   |